

Why Color Is Not Space, and the Spacetime Signature of D_{IV}^5 Is Matter-Induced

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Tier discipline up front. Part I is a theorem (an impossibility, DERIVED). Part II is a matter-induced boundary result built on 4 no-go theorems + 2 not-shown identifications — not “six closed routes”; a theorem cannot be “unmet,” and we do not inflate the count. Part III is a constructive narrative, near-complete, with each ingredient tiered honestly (framework where framework, open-program where open). Near-complete is not complete. One live test — the Hardy isometry — could promote Part III’s ontology from asserted to exhibited; we mark it as a test, not a result.

Abstract

General relativity, a century on, stands: spacetime is a Lorentzian (3,1) manifold and Lorentz invariance is the surviving symmetry. We ask whether that signature is derivable from the one geometry of Bubble Spacetime Theory, the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. Part I is a one-line impossibility any referee can check: the three spatial directions are not the three colors of $SU(3)$ — color is complex, space is real; $SU(3)$ has no real 3-dimensional representation, and BST’s reality operator J (charge conjugation) commutes with the color action on zero of 20,000 sampled group elements. The spatial three is instead the real Peirce component V_{12} , whose dimension $n_C - 2 = 3$ is forced by the Jordan structure — the Peirce route *explains* the three-ness of space, which no color route can. Part II: the global (3,1) signature is not forcible from D_{IV}^5 geometry and dynamics alone — four independent no-go theorems close the geometric routes; two further identifications remain not-shown — so the signature is matter-induced (relational, Machian), exactly as GR posits its manifold and lets matter curve it. Part III gives the constructive complement: a Markovian (3,1) spacetime — commitment classicalizes the process — with a UV-finite, fuzzy continuum whose resolution is proposed to be the Uncertainty Principle (an open program); its Lorentz-safety is a covariant construction with a matter-set frame (a spread of observer velocities — the matter-induced thesis again, GR’s own covariant-theory-with-a-CMB-frame situation), not an unqualified invariance. Confinement originates in D_{IV}^5 and is projected from the boundary at commitment; and commitment is the Born measurement — its record statistics reproduce the Born probabilities $|\psi|^2$ (classical information into the fermion record, *matter*) while the phase does not survive (quantum coherence removed, *the measurement*); the arrow of time is the *same* commit — already derived (T2553), and here shown to be one fact with the measurement, counted once.

Part I — The reality obstruction: color is not space (THEOREM)

The most seductive shortcut in the program is “three colors = three spatial directions.” As a statement about $SU(3)$ it is false, and the impossibility is one line.

- $SU(3)$ has no real 3-irrep (Elie 5281). The fundamental 3 has Frobenius–Schur indicator 0 ($3 \neq \bar{3}$, complex); the smallest real irrep of $SU(3)$ is the adjoint 8. So “color $SU(3)$ acting on three real spatial directions” is not an available group action. What acts on a real 3-space is $SO(3)$.
- The reality operator J cannot rescue it. BST’s spectral-triple real structure J is charge conjugation (color $\rightarrow$ anticolor — the operator that builds the Yukawa/quark-mass sector), an antilinear map $3 \rightarrow \bar{3}$. It is not a real structure *on* the 3 (that needs an invariant $\bar{3} \cong 3 = FS \neq 0$, false). Numerically, over 20,000 Haar-sampled $SU(3)$ elements, J commutes with the color action on exactly zero. Selecting a real $SO(3) \subset SU(3)$ would break the exact, unbroken color gauge symmetry — physically forbidden — and J , being color-equivariant, cannot pick a point in the 5-parameter family $SU(3)/SO(3)$. The quaternionic back-door ($Spin(5)=Sp(2)$, $J^2=-1$) is shut on the KO-dimension sign: a real $SO(3)$ needs $J^2=+1$.

- What “three colors are three directions” correctly means. The spatial three is V_{12} , the real off-diagonal Peirce component of the Jordan structure, with $SO(V_{12}) = SO(3)$; its dimension $n_C - 2 = 3$ is forced by the Peirce decomposition (T2545, “ V_{12} (real- $SO(3)$ -vector) irreducibility”). The Peirce route explains the three-ness of space. The color $SU(3)$ is a *separate, complex* structure that merely shares the integer 3 (the #35 discipline: never identify objects by a shared integer).

Consequence. Color has nothing to do with the structure of spacetime. This is a theorem, it leads the paper, and it retroactively explains why every color→space route we tried failed.

Part II — The (3,1) signature is matter-induced (T2565)

To *derive* the descent from the 5-dimensional substrate ($R \times S^4$) to observed (3,1) spacetime, the geometry+dynamics must supply a distinguished direction and a velocity/scale. It supplies neither. The honest tally is four no-go theorems plus two not-shown identifications — not “six closed routes.”

Four no-go theorems (DERIVED): 1. No preferred direction (5257). No single-valued $SO(5)$ -equivariant map with invariant inputs yields a distinguished R^5 direction — the only $SO(5)$ -fixed vector is 0 (Schur). 2. No motion (5273). The commit generator H_B is the central Casimir of $K=SO(5) \times SO(2)$; $\exp(-\tau H_B)$ commutes with $SO(5)$ ($\| [H_B, L_{ab}] \| = 7.5 \times 10^{-14}$). A Casimir multiplies; it does not push — the induced velocity is exactly 0. 3. No scale (5276). The commit-induced causal order is conformal, scale-free — it fixes $\Delta\theta/\Delta t \leq 1$ but never a length. 4. Color is not space (5281, Part I). The reality obstruction closes the color→space route as a theorem.

Two not-shown identifications (open, not theorems and not claimed as closed): - Confinement identification (§513): three indivisible color commitments $\leftrightarrow$ three spatial directions — not shown (and Part I shows the $SU(3)$ reading is unavailable; a non-gauge reading is unproven). - The corpus $4 \rightarrow 3$ chain: $SO(5,2) \rightarrow SO(4,2) \rightarrow SO(3,1)$ — either acts on the spinor/record index (γ^5 , the T2555 phase-drop, not a spatial dimension) or is Casimir-forbidden (theorem 2); not shown to reduce a spatial dimension.

Conclusion (DERIVED, negative). The $5 \rightarrow (3,1)$ descent is not forcible from D_{IV}^5 geometry+dynamics alone; it requires a matter/observer input. The signature is relational/Machian — matter induces it — a *located boundary*, not a failure, exactly as GR posits its manifold. Wording binding: “geometry-alone,” never “exhaustive” — the self-consistency route (Part III) remains open.

Hypotheses stated. (i) Theorem 2’s “no motion” is about the central Casimir specifically; the commit *projection* (Born-localization, T2542) supplies the per-observer *position* Ω_0 — so position is corpus-supplied per-observer, and the missing matter inputs are velocity and the site-measure. (ii) Theorem 1’s scope is single-valued equivariant inputs, not a closure over all constructions. (iii) SSB remains unsupported, not excluded.

Part III — The constructive narrative (near-complete, honestly tiered)

GR is correct. BST does not re-derive (3,1) from geometry — Parts I–II prove it cannot — but gives the honest construction of the spacetime matter induces.

1. A Markovian (3,1) spacetime — commitment classicalizes the process (framework tier)

The commit process is memoryless, and this is a genuine finding, not a semigroup tautology: uncommitted, Chapman–Kolmogorov fails (amplitudes interfere); committed, it holds exactly (Elie). Commitment is the Markovianization — this is decoherence, the quantum→classical boundary. The committed (classical) spacetime is the causal set with a derived order (T2564); the order relation is frame-invariant (5276), commit-energy is a time function, boosts are re-foliations. Whether the discrete *event structure* is Lorentz-invariant is not automatic and not claimed unqualified — Part III.3 gives the honest statement: a covariant construction (per-worldline proper-time tick) whose frame is matter-set (a spread of observer velocities), GR’s own covariant-theory-with-a-CMB-frame situation. (*Framework tier.*)

2. The spatial frame is V_{12} — and its three-ness is explained (calibrated up)

Three distinct objects, never blurred: - $\dim V_{12} = n_C - 2 = 3$ is forced by the Peirce decomposition — this explains the three-ness of space (the Peirce route, unlike color, is an explanation, not an identification). DERIVED. - The tangent (3,1) signature is Structure-Derived (T2545). - The $4 \rightarrow 3$ sky-reduction (dropping the radial line-of-sight) is a *different* object — matter-induced (T2565), and provably not geometry-forced.

3. The continuum is UV-finite and fuzzy — an OPEN PROGRAM

BST is UV-finite: $N_{\max} = 137$, GF(128), sums not integrals (F865). There is no $N \rightarrow \infty$ limit — which is why a continuum-limit dimension measurement was elusive; the dimension is set by the Peirce algebra (V_{12}), not by a limit. The proposal: the “continuum” is the finite-resolution observer’s coarse-grained view of the discrete Markov process, its resolution floor the Uncertainty Principle (Casey), “zero spacing” = sub-resolution. This is an open program, not banked — it requires three things, and we state exactly what is and isn’t done: 1. Exhibit the coarse-graining map — the τ -window is real and widens with the cap, but has only been checked on a torus Laplacian ($\sim 3\%$ at cap 137), *not BST’s real operator*. No window number goes in the paper until it is on the real spectrum. 2. The Lorentz requirement — covariant construction with a matter-set frame. A *discrete* structure is Lorentz-invariant only as a Poisson sprinkling (Bombelli–Henson–Sorkin); a *global* commitment tick — a preferred simultaneity layer — breaks Lorentz with perfect contrast (Elie 5284: $S = 1.0000$ in one frame, sharpening at large scales, not washing out). BST’s tick is instead per-worldline proper time — a covariant construction (T2564: boosts = re-foliations; but a re-foliation is a change of *description* — compatible with, not *forcing*, the local reading). What makes it Lorentz-safe is a matter condition (one-line hypothesis, stated not corpus-forced): a spread of observer velocities — no privileged velocity — at which the frame-contrast collapses (Elie 5284: $S \leq 0.006$). At *zero* spread the violation returns full-strength, so this is a genuine assumption — but it is exactly the Machian, matter-induced thesis of Part II (matter sets the frame), and it is physically free (~ 30 orders of margin). This is GR’s own situation: a covariant theory with a matter-set (CMB-like) frame — weaker than an unqualified “Lorentz-invariant,” and sufficient; BHS does not apply. The last continuum hole reduces to the lead thesis: matter commits. (The $N_{\max}$ cap is Lorentz-fine — a fixed count reads Poisson-like at every boost.) 3. Recover one continuum equation. Until all three land, the continuum reframe is a candidate.

4. Confinement from the boundary; and the commit = measurement + arrow (the exhibited spine)

Confinement originates in D_{IV}^5 and is projected from the boundary upon commitment (Casey) — the indivisibility of the committed record, a genuine substrate feature, *separate* from spacetime structure (Part I).

The interior $\leftrightarrow$ boundary ontology is now precise, and it exhibits both the measurement and the arrow of time from a single act: > Commitment is the Born measurement. Per event, a commit records *one* outcome (not a probability); across many commits the record statistics reproduce the Born probabilities $|\psi|^2$ — classical information written into the fermion record (T2543), *matter* — while the phase does not survive — quantum coherence removed, *the measurement* (Elie’s Chapman–Kolmogorov / decoherence finding). The arrow of time is the *same* commit, and it is already derived (T2553: H_B unbounded above $\Rightarrow$ no $\tau < 0 \Rightarrow$ pure positive-time; and the long-root Szegő projection whose kernel is the arrow). On the Hardy side this positivity reads as $n \geq 0$ (holomorphic, one-sided; Elie 5283, $|Q|=1$ on the Shilov boundary to 7.9×10^{-16}) — but that is the arrow *by definition* of the Hardy state space, not a fresh derivation: Hardy $n \geq 0$, $H_B \geq 0$, and Szegő irreversibility are one positivity in three languages, counted once (F1010). The content is the reconciliation — that measurement and the *banked* arrow are the *same single commit*, not two facts.

Two honest caveats, stated (not hidden). (i) The Szegő isometry is a classical theorem (Korányi–Stein–Szegő = $c_{FK}/T2442$) — we report it as a check of our implementation, not a discovery; and an isometry is *not* itself “information conservation” (a phase-drop is norm-preserving but coherence-losing), so the information is the Born statistics $|\psi|^2$, not the norm — the *reconciliation* is the content, not the isometry. (ii) The commit dynamics runs on the Bergman space (T2401); the Bergman $\rightarrow$ boundary map is an invertible isomorphism (no information lost) that is isometric in the Hardy norm but rescales blockwise in the Bergman norm. Part III leans only on what is exact — $|Q|=1$ on the boundary, k -independence of the boundary norm, blockwise orthogonality, and $n \geq 0$ on the image — and states the blockwise rescaling rather than glossing it. (*Verified: Elie 5283.*)

The one open frontier (marked, not hidden)

The self-consistency fixed point: matter is commitments (T2543), commitments make the order (T2564), the order constrains where commitments sit — a knot that must close on itself. If unique with a non-degenerate (height, not ordering-fraction) dimension read, “matter induces the signature” becomes “matter predicts it.” Existence is plausible; uniqueness is open — the sole route Part II does not close.

Summary table (honest tiers)

Piece	Status
Color-SU(3) $\neq$ spatial-3 (reality obstruction)	THEOREM (Part I)
$\dim V_{12} = n_C - 2 = 3$, spatial frame = V_{12} (real SO(3))	DERIVED — and <i>explains</i> the 3 (Peirce)
(3,1) tangent signature	DERIVED (T2545)
Markovian commit process = causal order;	FRAMEWORK (Chapman–Kolmogorov: fails
commitment classicalizes	uncommitted, holds committed)
(3,1) not forcible from geometry+dynamics	DERIVED negative (4 theorems + 2 not-shown)
$4 \rightarrow 3$ drop-radial = finite-resolution observer	MATTER-INDUCED (T2565); not geometry-forced
	(proven)
Continuum = UV-finite fuzzy coarse-graining,	OPEN PROGRAM (coarse-graining map + one
resolution = UP	continuum equation owed); Lorentz: covariant
	construction + matter-set frame (Elie 5284 — a matter
	hypothesis = the lead thesis, not corpus-forced; BHS
	doesn't apply)
Commit = Born: record statistics reproduce $ \psi ^2$ (one	DERIVED (arrow = T2553, counted once) +
outcome per event), phase dropped = measurement;	coherence reconciliation (F1010); isometry = classical
arrow = same commit	check
Self-consistency fixed point $\Rightarrow$ unique (3,1)	OPEN (would lift to matter-predicted)

The landing, stated plainly

A publishable pair: a one-line impossibility theorem leading (color is not space, checkable by any referee), a matter-induced boundary result behind it (the signature is relational, provably not geometric — four no-go theorems, two honest not-showns), and a constructive Markovian (3,1) narrative — spatial frame V_{12} (three-ness *explained* by Peirce), a UV-finite fuzzy continuum whose resolution is the Uncertainty Principle and whose Lorentz-safety is a covariant construction with a matter-set frame (GR's own situation, reducing to the lead thesis), and a single commit that is at once the measurement and the (already-derived, T2553) arrow of time. GR is correct; BST's addition is to prove *why* the signature must be relational — the substrate origin of “matter curves space-time.” No load-bearing computation remains. One lever would still upgrade the result from *matter-induced* to *matter-predicted*: the self-consistency knot (whether the commitment fixed point uniquely fixes the dimension, read by height, not the shape-degenerate ordering fraction). It is marked open; nothing beyond the stated tiers is claimed. Two days killed every pretty version until a true, humbler one stood.

Files: T2565 (registry); F997–F1010 (arc); T2545 (V_{12} real-SO(3)-vector / tangent (3,1)); T2564 (causal set, boosts = re-foliations); T2553 (positive-time arrow, $H_B \geq 0$); T2549/T2550/T2547 (real structure J); F865 (UV-finite); T2542/T2555 (commit projection); F1010 (arrow = holomorphic positivity, three languages counted once); the Hardy/Szegő machinery (Tier-0 H^2). Elie 5257/5273/5276/5281 (the four no-go theorems), 5283 (Hardy $n \geq 0$ / commit=Born reconciliation), 5284 (BAR-3: global tick breaks Lorentz, local proper-time tick safe). Nothing pushed; CP existence-only.